

- 3 Fig. 3.1 is a photomicrograph of human blood cells from a healthy individual who lives at sea level. The cells labelled **C**, **D** and **E** are white blood cells.

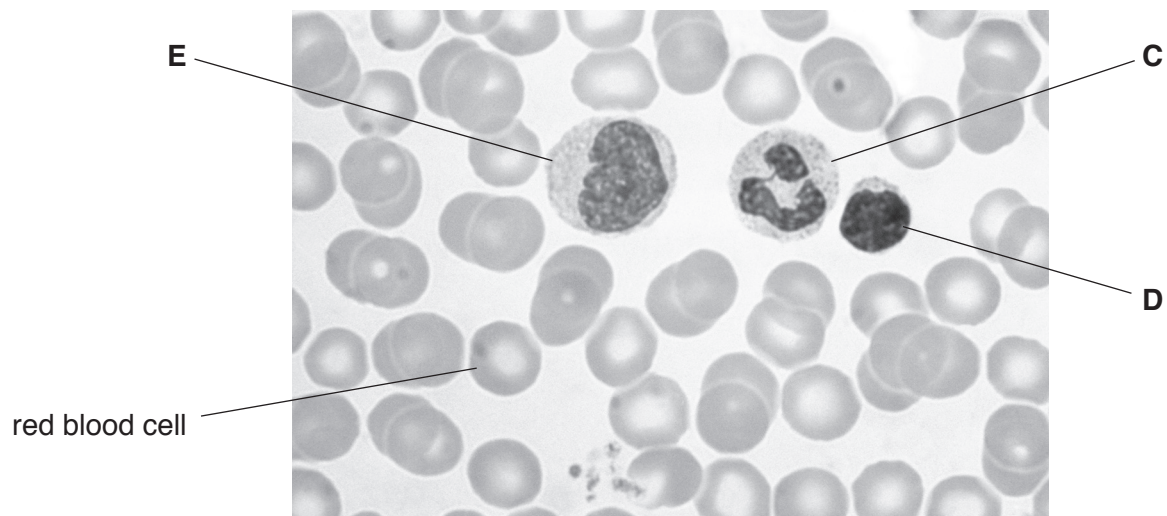


Fig. 3.1

- (a) Name cells **C**, **D** and **E**.

C

D

E

[3]

- (b) In humans, an increase in the white blood cell count can be associated with leukaemias and with infectious diseases, such as measles.

Chronic lymphocytic leukaemia (CLL) is a type of cancer that starts in the bone marrow. In the early stages, many people with CLL feel well. The disease is sometimes diagnosed by chance during a routine blood analysis, when a high white blood cell count is noticed. Many of these white blood cells are only partially mature.

- (i) Suggest why CLL starts in the bone marrow and **not** in any other location in the body.

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 [2]

- (ii) Explain why a high white blood cell count is a feature of measles and of CLL.

measles

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CLL

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..... [3]

- (c) Most of the oxygen that enters the mammalian circulatory system is transported by red blood cells.

- (i) Describe **and** explain the passage of oxygen across the cell surface membrane of the red blood cell.

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- (ii) At a high altitude, the partial pressure of oxygen in the atmosphere is lower than at sea level. If a person travels from low altitude to high altitude and remains there for a few weeks, the red blood cell count increases.

Explain why the body needs to respond to high altitude by increasing the number of red blood cells.

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(d) Polypeptide synthesis occurs before a red blood cell is released into the circulation.

The *HBB* gene codes for the β -globin polypeptide of haemoglobin.

There are two alleles of *HBB*, known as *Hb^A* and *Hb^S*.

Describe the difference between the *Hb^A* allele and the *Hb^S* allele **and** state how this difference affects:

- the β -globin polypeptide
- the haemoglobin molecule.

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[Total: 17]